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DS 120 Fall'25

Time: 50 minutes

Midterm 2

50 Points

Write down the steps involved for each question clearly. There will be partial credits for the correct legible steps. DO NOT write on the back side of the pages.

1. Let $V = \{(x, y) \in \mathbb{R}^2 : x + y = 0\}$ be a subset of $\mathbb{R}^2$ with standard vector addition and scalar multiplication. Is this a valid vector space? **5 points**

Hint: Check for presence of 0 vector and closure under standard vector addition and scalar multiplication.

$$\vec{v} = (0, 0) \rightarrow x + y = 0 \rightarrow 0 + 0 = 0 = 0$$

yes it is a valid vector space.

2. Let

$$\vec{v}_1 = \{1, 2, 3\}$$

$$\vec{v}_2 = \{2, 4, 6\}$$

$$\vec{v}_3 = \{0, 1, 1\}$$

- Is $\vec{v}_3$ in $\text{span}\{\vec{v}_1, \vec{v}_2\}$ **2 points**
- Are $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent? Why/Why not? **3 points**

$$\alpha_1(1, 2, 3) + \alpha_2(2, 4, 6) = (0, 1, 1)$$

$$(a_1 + 2a_2, 2a_1 + 4a_2, 3a_1 + 6a_2) = (0, 1, 1)$$

$$\begin{cases} a_1 + 2a_2 = 0 \\ 2(a_1 + 2a_2) = 1 \\ 3(a_1 + 2a_2) = 1 \end{cases} \rightarrow \begin{cases} a_1 + 2a_2 = 0 \\ a_1 + 2a_2 = 1/2 \rightarrow 0 \neq 1/2 \neq 1/3 \\ a_1 + 2a_2 = 1/3 \end{cases} \downarrow \vec{v}_3 \text{ not in } \text{span}(\vec{v}_1, \vec{v}_2)$$

$\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ are not linearly independent because for that to happen, at least one of the vectors must be the sum of the other 2, whereas this is not the case.

3. Let

$$\vec{v}_1 = \{1, 0, 1\}$$

$$\vec{v}_2 = \{0, 1, 1\}$$

$$\vec{v}_3 = \{1, 1, 2\}$$

Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ the basis for $\mathbb{R}^3$? Show your work and reasoning. **5 points**

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \rightarrow \text{no, } \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$$

isn't the basis for $\mathbb{R}^3$ as Identity matrix cannot be reached.

4. If the determinant of a 3×3 matrix A is zero, what can you say about: **3 points**

- The invertibility of A ? $\rightarrow$ not invertible
- The rank of A ? $\rightarrow 3$
- The system $A\mathbf{x} = \mathbf{b}$?

5. If two rows of a matrix are identical, what is its determinant? **2 points**

$$\text{Let } A = \begin{bmatrix} a & b \\ a & b \end{bmatrix} \rightarrow \det(A) = a \cdot b - a \cdot b \\ = \underline{\underline{0}}$$

6. Find the projection of $\vec{u} = (3,1)$ onto the line spanned by $\vec{v} = (2,1)$. 5

points

$$\text{Proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{7}{10} \cdot \vec{v}$$

$$= \frac{7}{10} (3,1) = \left(\frac{21}{10}, \frac{7}{10} \right)$$

7.

- Suppose we have the following data and we want to fit it into the line:

$$y = ax_1 + bx_2 + c$$

Handwritten note: c is y-intercept

observation	x_1	x_2	y
	2	1	5
	0	3	4
	4	2	10

$$\begin{bmatrix} 2a & b & | & 5 \\ 0 & 3b & | & 4 \\ 4a & 2b & | & 10 \end{bmatrix}$$

Find the coefficient matrix **3 points**

- You have a dataset with 10 observations and 10 predictors (plus intercept). What is the order of the coefficient matrix, A. What is the order of $A^T A$? **2 points**

8. Let

$$h(x, y) = x^3 - 3xy + y^3.$$

Show that the origin is a critical point that is a saddle point. **5 points**

$$\begin{aligned} h_x &= 3x^2 - 3y \\ h_y &= 3y^2 - 3x \end{aligned} \rightarrow 0 \text{ at } (1, 1) \text{ or } (0, 0)$$

critical points

$$\left. \begin{aligned} h_{xx} &= 6x \\ h_{yy} &= 6y \\ h_{xy} &= -3 \\ h_{yx} &= -3 \end{aligned} \right\} \text{hessian matrix} = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}$$

$$D = h_{xx} \cdot h_{yy} - (h_{xy})^2 = \underline{\underline{-6}} < 0 \rightarrow \underline{\underline{\text{saddle point}}}.$$

9. Temperature at a point in space is given by

$$T(x, y) = 3x^2y - y^3.$$

A bug is at point $(1, 2)$ and starts crawling in the direction of vector $\mathbf{v} = (3, 4)$.

Find:

- The **rate of change of temperature** in that direction. **6 points**
- The **direction** in which temperature increases fastest. What is the fastest rate of change in this direction? **4 points**

$$\begin{aligned} T_x &= 6xy \\ T_y &= 3x^2 - 3y^2 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{rate of change of temp in direction } \vec{v} = (3, 4) \\ \hookrightarrow (72, -21) \end{array}$$

10. A particle moves with a velocity:

$$v(t) = 3t^5 + 10t$$

in m/s. Find the **distance traveled** from $t = 0$ to $t = 4$ seconds. **5 points**

$$\int_0^4 3t^5 + 10t \, dt = \frac{3t^6}{6} + 5t^2 \bigg|_0^4 = 2048 + 80 = 2128 \text{ meters traveled.}$$

Extra work: DO NOT tear this page

Mention the question number next to the work.